

Hydrogen-station digital twin: separate evidence for separate claims

Public data → deterministic physics → bounded consequence results → expert-reviewed advice

Independent evidence

- **36 fast-fill tests**
Pressure · temperature · SOC
- **HyRAM+ v6.1**
Exact source + upstream tests
- **34 HIAD events**
Response fields withheld



Digital-twin evidence gates

1 Type-IV tank physics

Measured boundary conditions

2 Complete process loop

Controller · cascade · precooler

3 Consequence + SAGA

Adapter parity · blinded review



What the evidence currently supports

Tank validation: supported

RMSE 3.905 MPa · 4.694 °C · 3.812 %p

Full loop: screens failed

0/8 development · 0/11 internal comparison

HyRAM parity

47 tests · 803 subtests

SAGA pending

External expert review

Validated research components ≠ certified station control or autonomous emergency management